

Processing primary gene expression data: Normalization, harmonization and data quality control

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7.1 Background

Gene expression data are widely used in the fields of functional genomics and molecular medicine, e.g., in cancer research (slightly less than 500,000 PubMed papers found using search terms *gene expression* and *cancer* in May 2024). Two major approaches are used nowadays for large-scale transcriptional profiling: microarray hybridization (MH) of mRNA (Bednár, 2000; King and Sinha, 2001; Lashkari et al., 1997; Rew, 2001) and mRNA sequencing (RNAseq) (Chu and Corey, 2012; Ingolia et al., 2012; Korir et al., 2015; Maher et al., 2009; Nagalakshmi et al., 2008; Taylor et al., 2016; Wang et al., 2009). Both approaches utilize different rationales and can be further subdivided in several technological platforms. Consequently, the output data from MH and RNAseq are essentially platform-biased and require specific normalization/harmonization procedures in case any intercomparison may be needed.

The digital ocean of whole-transcriptome gene expression profiles has expanded since the early 2000s, when the first generation of robust and reproducible mRNA microarray hybridization (MH) techniques was introduced into the routine laboratory practice (Bednár, 2000; King and Sinha, 2001; Lashkari et al., 1997; Rew, 2001). The outstandingly high importance of the open-access gene expression data that could be accumulated and extracted from public databases was recognized immediately, thus leading to emergence of popular online repositories such as Gene Expression Omnibus (GEO) (Brazma et al., 2001; Edgar et al., 2002) or

ArrayExpress (Parkinson et al., 2007; Rocca-Serra et al., 2003). Later, this has also inspired many impactful large-scale integrative biomedical cooperative projects such as The Cancer Genome Atlas (TCGA) (The Cancer Genome Atlas Research Network, 2008; Tomczak et al., 2015) for cancer genomics and transcriptomics; Gene-Tissue Expression (GTEx) (GTEx Consortium, 2013; The GTEx Consortium et al., 2015) and Atlas of Normal Tissue Expression (ANTE) (Suntsova et al., 2019) for normal human tissue expression profiles; the CancerRxGene database (Yang et al., 2013) for genomes and transcriptomes of cell lines connected with their response to hundreds of drugs, the Broad Institute deconvoluted profiles for gene expression changes in cells under the influence of gene constructs, drugs, and other chemicals (Chen et al., 2016; Subramanian et al., 2007).

Shortly after the critical mass of gene expression profiles had accumulated, the following two conceptual problems with the data analysis were recognized. First is the poor technical compatibility of the expression profiles obtained using different experimental platforms/equipment and/or protocols and reagents (Chen et al., 2007; Liang, 2007; Mane et al., 2009; Shi et al., 2006; Wen et al., 2010). Indeed, this can be readily explained by the different physico-chemical principles of gene detection and interrogation (Han et al., 2021; Stelpflug et al., 2016) and by specific library preparation enzymatic bias (Tanaka et al., 2020). The second problem (so-called batch effect) dealt and still deals with the unclear compatibility of gene expression profiles obtained with the same equipment and reagents, but in different series of experiments, e.g., done in different time or in different labs (Demetrashvili et al., 2010; Lazar et al., 2013). There is no clear explanation of the nature of the batch effect (e.g., may be due to relatively different activities of enzymes and chemicals for library preparation and MH or RNA sequencing from batch to batch), but the effect itself is sound and frequently inevitable (Demetrashvili et al., 2010).

Many aspects of such intra- and cross-platform normalization of MH gene expression profiles were studied in the recent 2 decades. Namely, the approaches were formulated for studying the incomparability of profiles obtained using different platforms (Chen et al., 2007; Liang, 2007; Shi et al., 2006), for normalization of their expression data (Benito et al., 2004; Bolstad et al., 2003; Huang et al., 2012; Jiang et al., 2004; Johnson et al., 2007; Marron et al., 2007; Martinez et al., 2008; Shabalin et al., 2008; Warnat et al., 2005; Xia et al., 2009), and for assessing quality of their conormalization (Chen et al., 2007; Deshwar and Morris, 2014; Liang, 2007; Rudy and Valafar, 2011; Shi et al., 2006).

The compromised compatibility of gene expression profiles obtained using different platforms and protocols was experimentally explored in the international projects MAQC (for MH) and SEQC (for RNA sequencing). Both the MAQC (Chen et al., 2007; Liang, 2007; Shi et al., 2006) and SEQC (Xu et al., 2016) projects investigated the compatibilities of gene expression profiles obtained using various microarray and sequencing platforms for the same set of four sample types (named A, B, C, D), each done in multiple replicates. Type A samples were the commercially available Stratagene Universal Human Reference RNA specimens for all but brain human tissues; type B samples were also commercially available the Ambion Human Brain Reference RNA. Type C and D samples were the mixtures of A and B with A:B ratios of 3:1 and 1:3, respectively. In the MAQC project (Chen et al., 2007; Liang, 2007; Shi et al., 2006), the samples of types A,B,C, and D were profiled using the MH platforms Agilent-012,391 Whole Human Genome Oligo Microarray G4112A (GPL1708), Affymetrix Human Genome U133 Plus 2.0 Array (GPL570), and Illumina Sentrix

Human-6 Expression Beadchip (GPL2507). In the SEQC project (Xu et al., 2016), the same samples were profiled using the NGS platform Illumina HiSeq 2000 (GPL11154), as well as three MH platforms: Illumina HumanHT-12 V4.0 expression beadchip (GPL10558), Affymetrix Human Gene 2.0 ST Array (GPL17930), and Affymetrix GeneChip PrimeView Human Gene Expression Array (GPL16043).

The MAQC and SEQC projects investigated the correlations between the transcriptome profiles of the same biological type, yet obtained using the different experimental platforms. Although these correlations were high (Chen et al., 2007; Liang, 2007; Shi et al., 2006; Xu et al., 2016), without the some special cross-platform normalization methods (quantile normalization (Bolstad et al., 2003) was not enough), the overall collections of profiles were grouped according to the experimental platforms, rather than to the biological type of samples, both in terms of clustering dendrograms, and of principal component analysis (PCA) (Borisov et al., 2019; Deshwar and Morris, 2014; Junet et al., 2021; Rudy and Valafar, 2011; Shabalin et al., 2008).

As the reaction of the scientific community, a bunch of first-generation harmonization/normalization methods was generated in the first decade of 21st century, aimed at the standardizing of multiplatform expression profiles using specific algorithms. These methods were mostly trained on the different types of MH gene expression data and could dynamically transform gene profiles into a flexible yet intercomparable form (Carter et al., 2005). The following alternative approaches that had different principles and different destiny could be mentioned in this review: quantile normalization (QN) (Bolstad et al., 2003), quantile discretization (QD) (Warnat et al., 2005), normalized discretization (NorDi) (Martinez et al., 2008), distribution transformation (DisTran) (Jiang et al., 2004), empirical Bayes (EB)/ComBat (Johnson et al., 2007; Reynolds et al., 2023), distance-weighted discrimination (DWD) (Benito et al., 2004; Huang et al., 2012; Marron et al., 2007), cross-platform normalization (XPN) (Rudy and Valafar, 2011; Shabalin et al., 2008), gene quantiles (GQ) (Xia et al., 2009), and PPlatform-Independent Latent Dirichlet Allocation (PLIDA) (Deshwar and Morris, 2014).

Further approaches were largely influenced by the coming era of routine next-generation sequencing (NGS) of mRNA (RNA sequencing, or RNAseq) that has started roughly in the second decade of this century. Nowadays, RNAseq has become the gold standard and the basic tool for transcriptomic profiling (Chu and Corey, 2012; Ingolia et al., 2012; Korir et al., 2015; Maher et al., 2009; Nagalakshmi et al., 2008; Taylor et al., 2016; Wang et al., 2009). In addition to measuring gene activities, RNAseq has also potential of detecting mutations and overall tumor mutational burden (Hollern et al., 2019), gene splice isoforms (Thind et al., 2021), and oncogenic fusion transcripts (Gu et al., 2017; Kekeeva et al., 2016; Schmidt et al., 2018; Vellichirammal et al., 2020). During the RNAseq era, a new group of cross-platform data comparison methods was developed (Xu et al., 2016). However, the RNAseq gene expression profiles have outnumbered the MH counterparts relatively recently. It was only in 2019 (Borisov et al., 2019) when the number of datasets for the most popular RNAseq platform (Illumina HiSeq, 2000, GPL11154) exceeded the number of datasets for the most popular MH platform (Affymetrix U133, GPL570). Moreover, the total number of individual profiles for those two platforms is still comparable in 2024 as well. Many unique transcriptomic profiles exist only as the MH data, e.g., clinically annotated expression profiles for some pathological conditions including cancers (Borisov et al., 2020c).

For the RNAseq data, a method called *The Differential gene Expression in Sequencing*, DESeq (Anders and Huber, 2010)/DESeq2 (Love et al., 2014; Maza, 2016; Varet et al., 2016), based on the negative binomial distribution law, has rapidly become the standard in the field for the intraplatform normalization. However, an effective method for the cross-platform RNAseq, or for RNAseq versus MH harmonization, was missing until recently, although several attempts at simultaneous normalization of MH and RNAseq data must be mentioned, such as Training Distribution Machine (TDM) (Thompson et al., 2016), Universal exPression Code (UPC) (Piccolo et al., 2013), Feature-specific QN (FCQN) (Fauteux et al., 2021; Franks et al., 2018), MatchMixeR (MM) (Zhang et al., 2020), Integrative Bayesian Network (IBN) (Maleknia et al., 2020), Rank-in (Tang et al., 2021), and Elastic Shared LASSO Regularization (ESLR) (Huang et al., 2022) methods. The *divergence analysis* method is another interesting attempt to compare the MH and NGS mRNA expression profiles, as well as microRNA and DNA methylation data (Dinalankara et al., 2018). The researchers of the *divergence analysis* first applied conditional probability (Bayesian) models to mimic the unspecified (generalized-type) distribution that describes the gene expression/methylation data. This reconstruction was followed by the divergence analysis of one biological sample type from another. Although Dinalankara et al. (2018) managed to distinguish different samples by divergence analysis, the applicability of their approach to a wide range of popular MH and NGS platforms remains unexplored (Dinalankara et al., 2018).

Most of such crossplatform harmonization/normalization methods return the results in a flexible format. As such, the shape of the output normalized gene expression profiles fully depends on the group of samples under normalization and can be poorly compatible with the results of another normalization involving different transcriptional profiles. Thus, a new normalization procedure is typically required for every comparison.

Furthermore, taking into account next-order variables utilizing gene expression data, such as molecular pathway activation levels (PALs) (Buzdin et al., 2014; Borisov et al., 2017, 2020a,b, 2023a; Aliper et al., 2017; Ozerov et al., 2016), drug efficiency scores (DES) (Borisov et al., 2020a, 2023a; Poddubskaya et al., 2019; Tkachev et al., 2020b), and machine learning (ML) models (Borisov et al., 2018, 2020c, 2021a,b; Borisov and Buzdin, 2019; Tkachev et al., 2019, 2020a), this flexibility may unpredictably complicate gene expression analyses due to probable inconsistency of quantitative characteristic gene expression features.

More recently, a new concept was formulated for the harmonization methods: conversion of a whole set of profiles into the shape of a predefined experimental platform, e.g., in the Training Distribution Machine (TDM) method (Thompson et al., 2016). According to this paradigm, the harmonized results should look as if they were obtained using a single predefined gene expression profiling platform.

We then introduced a new type of uniformly shaped cross-platform harmonizers that use several mathematical transforms (Borisov et al., 2019, 2022, 2023b; Borisov and Buzdin, 2022). The first version of such software, Shambhala-1 (Borisov et al., 2019), used uniformly shaped harmonization that employed the piecewise-linear gene expression transformation method XPN (Rudy and Valafar, 2011; Shabalin et al., 2008). Later on, the XPN method was replaced in Shambhala by a more advanced piecewise-cubic method CuBlock (Junet et al., 2021), thus giving next version termed Shambhala-2 (Borisov et al., 2022, 2023b).

Current versions of Shambhala utilize transformation of a fraction of ~ 8000 most strongly expressed human genes because their transcriptional activities can be assessed with the

greatest precision compared to the low-expressed genes (Borisov et al., 2019, 2023b). In our report (Borisov et al., 2022) we showed that Shambhala-2 returns transformed gene expression profiles that are clustered according to their biological origin rather than by their experimental platform. However, it remained unexplored whether this harmonization also retains differential gene expression features that can functionally characterize the samples under analysis.

Later on (Borisov et al., 2023b), we classified available intra- and cross-platform harmonization methods of transcriptomic profiles and compare their performance characteristics. In this study (Borisov et al., 2023b), we tested Shambhala performance in retention of fold-change gene expression features and other functional characteristics of gene clusters such as pathway gene activation levels and predicted cancer drug activity scores. Using 6793 cancer and 11,135 normal tissue gene expression profiles from the literature and experimental datasets, we applied 12 performance criteria for different versions of Shambhala and other methods of transcriptomic harmonization with flexible output data format. Such criteria dealt with the biological type classifiers, hierarchical clustering, correlation/regression properties, and stability of drug efficiency scores. We also assessed the quality of Shambhala output data for building both local and global machine learning expression-based classifiers of human cancer and normal tissue types. The piecewise-cubic Shambhala-2 harmonizer demonstrated the best results with the close to one correlation and linear regression coefficients for the comparison of training versus validation datasets and more than two times lesser instability for calculation of drug efficiency scores compared to other methods.

Finally, we applied Shambhala-1/2 methods (that were initially invented for processing MH and bulk mRNA-seq data) to harmonization of single-cell-seq (sc-seq) Big Data (more than 200,000 profiles corresponding to more than 2000 different knockouts (KO) or retinal pigment epithelial, RPE1, cell lines (Replogle et al., 2022)). Again, the Shambhala-2 method applied after the initial Markov Affinity-based Graph Imputation of Cells (MAGIC) protocol (Andrews and Hemberg, 2018; Hou et al., 2020; Luo et al., 2020; Su et al., 2014; Xu et al., 2021) of sc-seq profiles showed the best performance in terms of recognition of cell cycle phases.

In this chapter, we classify available intra- and crossplatform harmonization methods of transcriptomic profiles and compare their performance characteristics. We also describe the advantages of Shambhala, particularly Shambhala-2, in various computational tests. At the end, we include practical recommendations that may guide the reader to select an optimal method depending on a specific task.

7.2 Principles of harmonization algorithms

Different harmonization methods are based on different algorithms aimed to suppress the platform bias and batch effect. These algorithms may utilize different approaches for gene expression data processing and produce output data in different formats. Considering the mathematical apparatus, we propose the following classification:

- (1) Methods based on statistical transformations (considering quantiles, ranks, means, medians of gene expression levels, *etc.*):
 - (a) those using ranking of expression levels and setting the output levels according to the averaged values, such as QN (Bolstad et al., 2003), Feature-Specific QN (FCQN)

- (Franks et al., 2018), Quantile Discretization (QD) (Warnat et al., 2005), Gene Quantiles (GQ) (Xia et al., 2009), Normalized Discretization (NorDi) (Martinez et al., 2008), Distribution Transformation (DisTran) (Jiang et al., 2004; Warnat et al., 2005), Median Rank Scores (MRS) (Warnat et al., 2005), YuGene (Lê Cao et al., 2014), and Rank-in (Tang et al., 2021);
- (b) those using piecewise rescaling of log-expression levels according to the mean/median values of distinct genes and samples, such as Column Sample (CS), Median-Centered (MC) (Shabalin et al., 2008), and Analysis of Variance (ANNOVA) (Nguyen et al., 2020) method.
- (2) Methods using regression and/or maximum likelihood models for validation of predefined statistical hypotheses:
- (a) those using negative binomial distribution, such as the DESeq (Anders and Huber, 2010)/DESeq2 (Love et al., 2014; Maza, 2016; Varet et al., 2016);
 - (b) those using log-normal distribution with either covariance analysis (Ou-Yang et al., 2017), or with conditional/Bayesian models, as for the methods Universal exPRession Code (UPC) (Piccolo et al., 2012, 2013), Empirical Bayes (ComBat) (Johnson et al., 2007; Reynolds et al., 2023), Robust Microarray Analysis (RMA) (Irizarry et al., 2003), GeneChip Robust Multiarray Analysis (gcRMA) (Wu et al., 2004), Model-Based Expression Indices (MBEI) (Li and Wong, 2001), Probe Logarithmic Intensity ERRor (PLIER) estimation (Therneau and Ballman, 2008), frozen Robust Microarray Analysis (fRMA) (McCall et al., 2010; 2011a,b; 2012), MatchMixeR (MM) (Zhang et al., 2020), Cross-Platform Comparison (XPC) (Zhang et al., 2021);
 - (c) those using Dirichlet and gamma distributions, as for the method Platform-Independent Latent Dirichlet Allocation (PLIDA) (Deshwar and Morris, 2014);
 - (d) those using the empirical superposition of conditional probabilistic (Bayesian) models that describe the generalized-type distribution, as for the method applied for the comparison of the MH, NGS, microRNA, and DNA methylation data (Dinalankara et al., 2018; Maleknia et al., 2020);
 - (e) those using the Least Absolute Shrinkage and Selection Operator (LASSO) regression models (Huang et al., 2022).
- (3) Methods finding similar clusters in gene expression matrices of the datasets under normalization, and then using iterative corrections to fit each cluster as close as possible to the target model:
- (a) those using piecewise-linear interpolations in the log-expression space, such as Cross-Platform Normalization (XPN) (Shabalin et al., 2008);
 - (b) those using piecewise-cubic interpolations in the log-expression space, such as CuBlock (Junet et al., 2021).
- (4) Methods utilizing machine learning (ML) to find and artificially remove dissimilarities between datasets to be normalized:
- (a) those using the linear *support vector machine* (SVM) ML method, such as Distance-Weighted Discrimination (DWD) (Benito et al., 2004; Huang et al., 2012; Marron et al., 2007);

- (b) those using quantile-based regression models for data transfer from source to target datasets, such as Training Distribution Machine (TDM) (Thompson et al., 2016).

Another important aspect that must be considered in this review is the format of output gene expression data generated by the harmonization techniques. Most currently existing methods return the results in a *flexible* format. For the flexible normalization, the shape of the output transformed gene expression profiles is a variable that depends on all the profiles under harmonization. This has an important limitation: one cannot combine the output datasets generated after two or more acts of such harmonization. Even adding as few as just one transcriptional profile would require a new harmonization of the entire dataset. This clearly increases the calculation costs for the large datasets that are being routinely updated.

Taken together, these factors complicate the analysis of not only single gene expression levels but also of higher order gene-based biomarkers such as gene signatures (Lee et al., 2021), molecular pathway activation levels (Borisov et al., 2020b), algorithmically deduced cancer drug efficiency scores (Poddubskaya et al., 2019; Tkachev et al., 2020b), and different ML models (Tkachev et al., 2019, 2020a; Turki and Wang, 2019).

To overcome these limitations, an alternative concept was formulated comprising conversion of a whole set of profiles under harmonization into a predefined output shape, e.g., into a shape of a preferred gene interrogating experimental platform. In such paradigm, the harmonized output should look as if it would be obtained using a predefined gene expression platform. The examples of predefined shape harmonization methods include Frozen Robust Microarray Analysis (fRMA) (McCall et al., 2010; 2011a,b; 2012), robust Quantile Normalization (Bolstad, 2008), Training Distribution Machine (TDM) (Thompson et al., 2016), and Universal exPression Code (UPC) (Piccolo et al., 2013).

More recently, we proposed a new family of uniformly shaped crossplatform harmonizers termed Shambhala (Borisov et al., 2019, 2022, 2023b). Harmonization here (see Fig. 7.1) is performed not simultaneously for all the profiles under harmonization, but for the gene expression profiles taken one by one, when each individual profile is merged and quantile-normalized (Bolstad et al., 2003) with an auxiliary calibration dataset that is predefined by the method developers. Then the resulting dataset is converted into the shape of the so-called *reference definitive dataset*. This creates an additional advantage of coharmonizing datasets of different, even noncomparable, sizes.

Furthermore, such harmonization may use different mathematical transforms as the engine to reshape the transcriptional profiles. The first version of Shambhala used the piecewise-linear method XPN (Rudy and Valafar, 2011; Shabalin et al., 2008) for profile reshaping (Borisov et al., 2019), whereas the latest version (Borisov et al., 2022, 2023b) utilized piecewise-cubic transformation method CuBlock (Junet et al., 2021).

7.3 Evaluation of the quality of harmonization

Harmonization of transcriptional profiles is a complex process that can distort functionally relevant features like clustering and neighborhood on a dendrogram, fold-change of gene

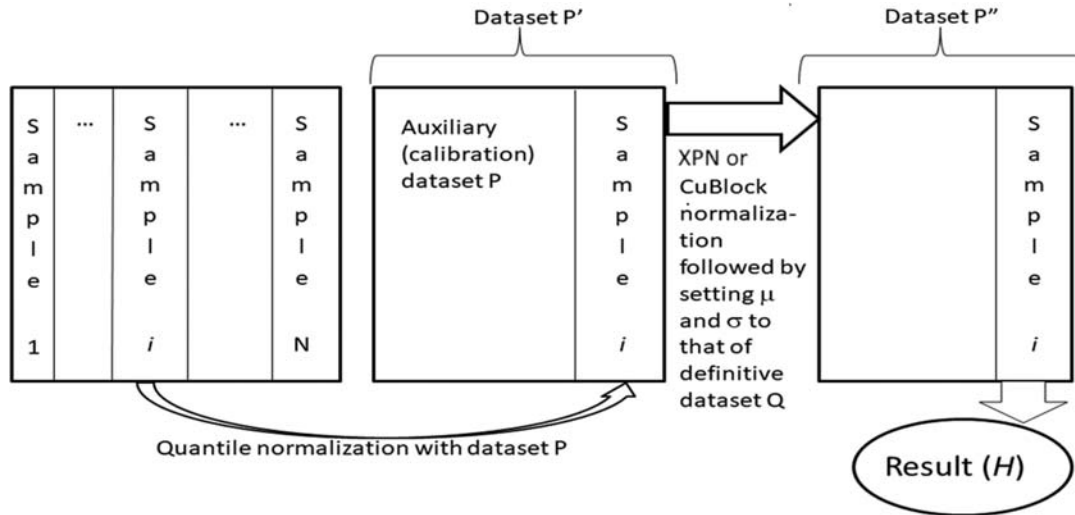


FIGURE 7.1 Shambhala-1/2 approach (Borisov et al., 2019, 2022, 2023b; Borisov and Buzdin, 2022) to uniformly shaped harmonization of gene expression data. Gene expression profiles for samples (1, ..., N), e.g., obtained using different experimental platforms are taken one by one, separately merged, and quantile-normalized with an auxiliary calibration dataset P . The resulting dataset P' is then transformed into the format of the reference definitive dataset Q , which results in dataset P'' . The latter contains the finally transformed profile of initial sample i , which is considered harmonized (H). The profiles of all other samples are harmonized one by one using the same algorithm.

expression with relation to control samples, etc. We listed in Table 7.1 some of the quality assessment metrics and the abilities of the different methods to retain the initial functional characteristics in the output profiles after harmonization.

The following quantitative metrics and methods may apply to estimate the effect of harmonization:

- (1) First, different statistical criteria may be used to estimate the following endpoints:
 - (a) correlation/regression analysis for the gene expression profiles before and after harmonization (Borisov et al., 2022, 2023b; Deshwar and Morris, 2014; Junet et al., 2021; Rudy and Valafar, 2011; Shabalin et al., 2008);
 - (b) comparison of between- and within-class distances before and after harmonization (Shabalin et al., 2008).
- (2) Alternatively, one may classify the samples according to gene expression data after normalization, involving various machine learning (ML) methods:
 - (a) logistic regression (Friedman et al., 2010), used in (Deshwar and Morris, 2014);
 - (b) SVM (Vapnik and Chapelle, 2000), used in (Borisov et al., 2023b; Rudy and Valafar, 2011; Shabalin et al., 2008);
 - (c) nearest shrunken centroids Prediction Analysis for Microarrays (PAM) (Tibshirani et al., 2002), used in (Shabalin et al., 2008);
 - (d) k nearest neighbors (kNN) method, used in (Borisov et al., 2023b).

TABLE 7.1 Selected benchmarks of harmonization methods.

Reference for comparison	Methods	Materials	Experimental platform	Qualitative criteria	Quantitative criteria	Best methods
Shabalin et al. (2008)	Crossplatform normalization, XPN (Shabalin et al., 2008); column sample (CS); Median-centered (MC); empirical Bayes (EB) (Johnson et al., 2007) Distance weighted discrimination (DWD) (Benito et al., 2004 ; Marron et al., 2007)	Three breast cancer datasets (Hu et al., 2006 ; Huang et al., 2003 ; van 't Veer et al., 2002)	Affymetrix GeneChip U95Av2 arrays (Huang et al., 2003); 25K Agilent oligonucleotide arrays (Hu et al., 2006 ; van 't Veer et al., 2002)	—	Average distance to the nearest sample in another Platform; correlation with column standardized data; global integrative correlation; Preservation of significantly differential genes	XPN
Rudy and Valafar (2011)	XPN; DWD; EB (ComBat) (Johnson et al., 2007 ; Reynolds et al., 2023); median rank scores (MRS) (Warnat et al., 2005); quantile discretization (QD) (Warnat et al., 2005); normalized discretization (NorDi) (Martinez et al., 2008); distribution transformation (DisTran) (Jiang et al., 2004 ; Warnat et al., 2005); gene quantiles (GQ) (Xia et al., 2009); quantile normalization (QN) (Bolstad et al., 2003)	MAQC dataset (Chen et al., 2007 ; Liang, 2007 ; Shi et al., 2006)	Human genome Survey microarray v2.0; Agilent-012391 whole human genome oligo microarray G4112A; Affymetrix human genome U133 plus 2.0 Array; Illumina Sentrix Human-6 expression beadchip	—	Mean - mean regression; crossdataset data transfer for linear SVM (Vapnik and Chapelle, 2000) and nearest shrunken centroids (Tibshirani et al., 2002) classification.	XPN (for datasets of comparable size); DWD (for datasets of noncomparable size)
Deshwar and Morris (2014)	XPN; DWD; platform-independent latent Dirichlet allocation (PLIDA) (Deshwar and Morris, 2014)	Prostate cancer datasets (Jia et al., 2011 ; Wang et al., 2010); Breast cancer datasets (Desmedt et al., 2007 ; Hu et al., 2006); MAQC	Affymetrix human genome U133 Array; Agilent human 1A (V2); Human genome Survey microarray v2.0; Agilent-012391 whole human genome Oligo microarray G4112A; Affymetrix human genome U133 plus 2.0 Array; Illumina Sentrix Human-6 expression beadchip	Visual inspection of PCA plots.	Correlation analysis between the profiles before and after normalization; Cross-dataset data transfer for logistic regression (Friedman et al., 2010) classification.	PLIDA

(Continued)

TABLE 7.1 Selected benchmarks of harmonization methods.—cont'd

Reference for comparison	Methods	Materials	Experimental platform	Qualitative criteria	Quantitative criteria	Best methods
Zhang et al. (2020)	MatchMixeR (MM) (Zhang et al., 2020); DWD; XPM; ComBat	NCI60 cell lines (dataset 1: 58 lines; dataset 2: 59 lines)	Affymetrix human genome U133A array; human genome U133 plus 2.0 Array; Agilent human genome whole microarray; Illumina HiSeq 2000	—	R ² score ^a analysis; F1 score ^b analysis	MM
Borisov et al. (2019)	Shambhala-1; QN; differential gene expression in sequencing 2 (DESeq2) (Love et al., 2014 ; Maza, 2016 ; Varet et al., 2016)	MAQC; SEQC datasets (Xu et al., 2016)	Agilent-012391 whole human genome oligo microarray G4112A; Affymetrix human genome U133 plus 2.0 Array; Illumina Sentrix Human-6 expression beadchip; Illumina HiSeq 2000; Illumina HumanHT-12 V4.0 expression beadchip; Affymetrix human gene 2.0 ST Array; Affymetrix GeneChip PrimeView human gene expression array	Visual inspection of cluster dendrograms	—	Shambhala-1 (linear Shambhala)
Junet et al. (2021)	CuBlock (Junet et al., 2021); ComBat (Johnson et al., 2007); YuGene (Lê Cao et al., 2014); DBNorm (Meng et al., 2017); Shambhala-1 (Borisov et al., 2019); universal exPression Code (UPC) (Piccolo et al., 2013).	MAQC	Agilent-012391 whole human genome Oligo microarray G4112A; Affymetrix human genome U133 plus 2.0 Array; Illumina Sentrix Human-6 expression beadchip	Visual inspection of cluster dendrograms and PCA plots.	Crossdataset data transfer for support vector machine (SVM) (Vapnik and Chapelle, 2000) classification.	CuBlock

Borisov et al. (2022)	Shambhala-2; Shambhala-1; QN; DESeq2; CuBlock; robust QN (QNR) (Bolstad, 2008); training distribution machine (TDM) (Thompson et al., 2016); UPC.	GTEx (GTEx Consortium, 2013), the cancer genome Atlas (TCGA) (Tomczak et al., 2015); Oncobox Atlas of normal tissue expression (ANTE) (Suntsova et al., 2019); MAQC; SEQC	Illumina HiSeq 2000; Illumina HiSeq 3000; Agilent-012391 whole human genome Oligo microarray G4112A; Affymetrix human genome U133 plus 2.0 Array; Illumina Sentrix Human-6 expression beadchip; Illumina HumanHT-12 V4.0 expression beadchip; Affymetrix human gene 2.0 ST array; Affymetrix GeneChip prime view human gene expression array	Visual inspection of PCA plots.	Watermelon multisection metric for quantitative assessment of clustering on dendrograms (Zolotovskaia et al., 2020)	Shambhala-2 (cubic Shambhala)
Borisov et al. (2023b)	Shambhala-2; Shambhala-1; QN; DESeq2; ComBat	Cancer TCGA; Cancer Oncobox; Normal TCGA. Normal Oncobox ANTE	Illumina HiSeq 2000; Illumina HiSeq 3000; Affymetrix human gene 1.1 ST array	—	ML accuracy, Watermelon Multisection metric; Correlation/regression analysis; sign-constancy analysis; retention of biological properties	Shambhala-2
Single-cell seq big data study (Repogle et al., 2022)	Shambhala-1, Shambhala-2, both e applied alone a and after the MAGIC imputation (Andrews and Hemberg, 2018; Hou et al., 2020; Luo et al., 2020; Su et al., 2014; Xu et al., 2021), RUV (Hung, 2019; Molania et al., 2023; Salim et al., 2022)	RPE1 cell lines (Repogle et al., 2022)	Illumina NovaSeq 6000 (sc-seq)		Correlation analysis of var-to-mean diagrams. Best distinguished between different phases of the cell cycle according to MW-metric	Shmabhala-2 prerun by MAGIC imputation

^a R^2 is the proportion of the variation in the dependent variable that is predictable from the independent variable Chicco et al, (2021).

^bF1 score is the harmonic mean of precision and recall Chicco (2017), Chicco and Jurman (2020).

As a typical material for such normalization quality benchmarks, in many studies the investigators used standardized reference samples, whose gene expression was interrogated with different equipment using different experimental protocols. Probably, the most important series of such crosscomparisons were performed within the microarray quality control (MAQC) (Chen et al., 2007; Liang, 2007; Shi et al., 2006) and sequencing quality control (SEQC) (Xu et al., 2016) projects mentioned above in this article.

The MAQC and SEQC projects were focused on profiling the specific model human mRNA sample types. One was the commercial Stratagene universal human reference RNA mixture for all but brain tissues; another one was the commercial Ambion human brain reference mRNA; and the two remaining types were the mixtures of the Stratagene/Ambion samples in the ratios 3:1 and 1:3, respectively.

The quality assessment is based on the expectation that a perfect harmonization must support the similarity of gene expression profiles according to the biological nature of the sample rather than depending on the equipment and reagents used to interrogate gene expression. Thus, early approaches used visual inspection of the principal component analysis (PCA) plots and/or cluster dendrograms to assess the crossplatform harmonization benchmarks (Borisov et al., 2019, 2022; Deshwar and Morris, 2014; Junet et al., 2021; Rudy and Valafar, 2011). However, this could only support a manual qualitative assessment without precise quantitative interrogation of the complex class distribution profiles.

We recently proposed a new metric for the algorithmic cluster analysis of dendrograms (Borisov et al., 2022, 2023b; Zolotovskaia et al., 2020) called watermelon multisection (WM). WM measures the strength of data matching with the trait of interest. When moving from the root of the dendrogram to its distal branches, one can calculate general decrease in entropy and, therefore, information gain (IG) at each node of the dendrogram, i.e., its split into two shoulders (Zolotovskaia et al., 2020). This accumulated and normalized IG constitutes the WM metric for a given dendrogram, and a given set of classes under analysis. Consequently, the ratio $= \frac{WM_S}{WM_P}$, where WM_S is WM metric for clustering according to classes corresponding to biological nature, and WM_P – according to the experimental platform used – may be used as a facile yet robust estimate of the harmonization quality. The higher R corresponds to the better quality, and vice versa (Borisov et al., 2022, 2023b).

7.4 Validation of Shambhala-1/2 protocol on bulk mRNA-seq profiles

Our validation (Borisov et al., 2023b) of the Shambhala method for uniformly shaped harmonization included the following tests (Fig. 7.2):

- (1) Comparison of different cancer and normal tissue datasets of gene expression profiles obtained using different equipment/protocols. The normalization methods included QN (Bolstad et al., 2003), DESeq2 (Love et al., 2014), empirical Bayes a.k.a ComBat (Johnson et al., 2007; Lagani et al., 2016), and different modifications of linear and cubic Shambhala. The quality control metrics included are:
 - (a) the accuracy of the transfer learning classifier, where the ML model is trained on profiles obtained in one batch and validated on profiles obtained in different batches and using different platforms/protocols;

A Test 1: Comparison of different cancer and normal tissue datasets of gene expression profiles obtained using different equipment/toolkits

Materials: one cancer and four normal tissue datasets
 Normalization methods: QN, DESeq2, ComBat, Shambhala
 Performance criteria: accuracy of transfer learning classifiers, closeness to 1 of correlation/regression coefficients between identical profiles, sample clustering according to biological type rather than experimental platform

B Test 2: Analysis of correlation and linear regression coefficients, as well as sign stability for downstream measures of case-to-control LFC, PAL and BES

Materials: cancer and corresponding normal tissue dataset
 Normalization methods: QN, Shambhala
 Performance criteria: closeness to 1 of correlation/regression coefficients between QN and Shambhala-obtained values, and between values with different control references, the minimal percentage for sign-changing events for each comparison

C Test 3: Retention of biologically relevant differences between various cancer profiles

Materials: cancer dataset for sex chromosome genes, breast cancer dataset
 Normalization methods: QN, DESeq2, Shambhala
 Performance criteria: Keeping biologically-grounded distinctions

FIGURE 7.2 Study design. (A) Comparison of different cancer and normal tissue datasets of gene expression profiles obtained using different equipment/toolkits. (B) Analysis of correlation and linear regression coefficients, as well as sign stability for downstream measures of gene expression derivatives. (C) Analysis for retention of biologically relevant differences between various cancer profiles after harmonization.

- (b) clustering quality of expression profiles after harmonization: “good harmonization” means that such clustering corresponds to the biological type of the sample rather than to batch or experimental platform;
 - (c) correlation and linear regression coefficient between expression profiles for the same tissue type, obtained using different equipment/protocols: should be close to one for a good harmonization.
- (2) Analysis of correlation and linear regression coefficients, as well as sign stability for downstream measures of gene expression derivatives (case-no-control log-fold-changes, pathway activation levels (PALs) (Aliper et al., 2017), and drug balanced efficiency scores (BESs) (Tkachev et al., 2020b). For a good harmonization, the correlation and linear regression coefficients should be close to 1, and the sign change rates should be as low as possible. The correlation, linear regression, and sign stability tests were performed under the following conditions:
- (a) different linear and cubic Shambhala modes versus QN;

- (b) LFC, PAL, and BES values calculated using Oncobox ANTE control sampling versus GTEx control sampling.
- (3) Retention of biologically relevant differences between various types of profiles after harmonization (QN, DESeq2, and multiple Shambhala modifications):
 - (a) differences between the male and female individual samples for the genes located on sex chromosomes,
 - (b) differences between cancer expression profiles in hormone-dependent, HER2-positive, and triple-negative breast cancer patients.

7.5 Differential clustering of human normal and cancer expression profiles

We first investigated the ability of Shambhala to support tissue-specific clustering of harmonized expression profiles in comparison with other nonuniformly shaped harmonization methods. We obtained human tissue gene expression datasets from the Gene-Tissue Expression (GTEx) repository ([GTEx Consortium, 2013](#); [The GTEx Consortium et al., 2015](#)), The Cancer Genome Atlas (TCGA) database ([Tomczak et al., 2015](#)), Atlas of Normal Tissue Expression (ANTE) collection of expression profiles ([Suntsova et al., 2019](#)), and from the Oncobox experimental collection of cancer tissue RNA sequencing profiles. The samples were obtained using the versions of Illumina NGS platform: HiSeq 2000 for the TCGA and GTEx RNA sequencing data, and HiSeq 3000 for the ANTE and Oncobox cancer data, and the microarray hybridization (MH) platform Affymetrix Human Gene 1.1 ST Array for the GTEx MH data. Taken together, they represented 37 human tissue types including 15 cancer and 22 normal tissue types. Four different gene library preparation protocols were used for obtaining these datasets, one common protocol for the ANTE and Oncobox data, one common for the normal and cancer TCGA data, and specific protocols for the GTEx RNA sequencing and MH data.

To compare performance, we conducted a harmonization procedure using the following alternative methods: (i) Quantile normalization, QN, a gold standard for normalization of MH gene expression data ([Bolstad et al., 2003](#)); (ii) DESeq2, a gold standard for normalizing RNAseq gene expression data ([Love et al., 2014](#); [Maza, 2016](#); [Varet et al., 2016](#)); (iii) ComBat, specially developed for batch effect elimination ([Johnson et al., 2007](#); [Lagani et al., 2016](#)); (iv) Shambhala-1, a uniformly shaped harmonization method with the XPN gene expression transformation module ([Borisov et al., 2019](#)), Sh1; (v) Shambhala-2 ([Borisov et al., 2022](#)), which is a uniformly shaped harmonization method with the CuBlock gene expression transformation module.

The P -, Q -, and R -rescaling modes for Shambhala-2 (Sh2PBR, Sh2QBR, and Sh2RBR, respectively) differ based on the mean and standard deviation values, which are applied to set the log-expression levels after the harmonization procedure ([Borisov et al., 2023b](#)). It is noteworthy that all modes of Shambhala reduced the number of genes after harmonization to ~ 8000 of the most strongly expressed “reaper” human genes ([Borisov et al., 2019](#)). The genes in the harmonized output are formed by the intersection of genes in three datasets: the raw (R) and two auxiliary datasets (P and Q). The best results, in terms of stressing the biological origin of the profiles and banning the artifacts generated by platform/protocol-

specific bias, were obtained for the P -dataset Affymetrix Human Genome U133A 2.0 Array (dataset P_0), which contained 8385 genes (Borisov et al., 2022). This MH platform is often used for routine transcriptomic profiling. Thus, using this P -dataset acts fairly similarly to the explicit expression filtering and increases the signal-to-noise ratio, thereby highlighting the biological origin of the samples. For comparability between different normalization methods, we performed all QN, DESeq2, and ComBat calculations with this set of highly expressed ~ 8000 genes, which helps determine the biological origin of the profile.

We then applied the following machine learning (ML) methods to build tissue type classifiers for the harmonized profiles: (i) 11 nearest neighbors; (ii) first nearest neighbor; (iii) linear SVMs. We performed 6 ML tests, one with cancer samples, and five with normal samples. In such tests, the training and validation groups of samples were taken from different initial datasets.

For each transfer learning test and each sample type in the classifier, we calculated the accuracy, i.e., the percentage of correct tissue type predictions in the validation dataset. Importantly, uniform output Shambhala method showed comparable performance with the gold standard flexible output methods QN and DESeq2, for both local (kNN-based) and global (SVM-based) classifiers (Figs. 7.3A,C and D and 7.4). Note that QN, DESeq2, and ComBat methods were also applied to the ~ 8000 genes, which survived Shambhala harmonization. Furthermore, the ComBat method, designed to remove batch effects, artificially makes two or more datasets under analysis to look similar; however, it mixes profiles with the different biological origins obtained using the same platform, which results in poor ML performance (Figs. 7.3B and 7.4).

We assessed the elimination of the platform/protocol-specific bias after harmonization using the statistical method *Watermelon Multisection* (WM) (Borisov et al., 2022, 2023b; Borisov and Buzdin, 2022; Zolotovskaia et al., 2020). WM enables tracking entropy loss/information gain at each node of the clustering dendrogram when moving in the direction from the root to the distal branches, thus giving WM metric for a given dendrogram. It can assess the quality of sample clustering according to known predefined groups (in our case, tissue types). The higher is WM metric, the better is the clustering according to known tissue types, and vice versa.

We, therefore, calculated WM metrics for the harmonized tissue samples in two major settings: WM_P for the classes corresponding to experimental platforms/protocols (e.g., TCGA-RNAseq, Oncobox-RNAseq, etc.) and WM_S for the classes corresponding to tissue types. Thus, the ratio $R = \frac{WM_S}{WM_P}$ may serve as the measure for harmonization quality. The higher the value of R is, the better is the clustering according to tissue type in relation to platform-specific bias, and vice versa.

In our analysis, the WM metric showed that the ComBat method had the best performance for platform bias elimination in most ML trials with cancer and normal human tissues, except when merging GTEx and TCGA normal RNAseq profiles and merging GTEx and ANTE normal RNAseq profiles. However, this elimination of the batch effects is only apparent and does not result in proper clustering of the same sample types. The high values of R for WM metrics are provided by low values of WM_P , rather than the high values of WM_S , and profile clustering after ComBat is not done according to the tissue type (Borisov et al., 2023b). Thus, according to our findings, ComBat method did not demonstrate an overall superior output as it blurs the similarity between the profiles of the same biological type, even when obtained using one experimental platform.

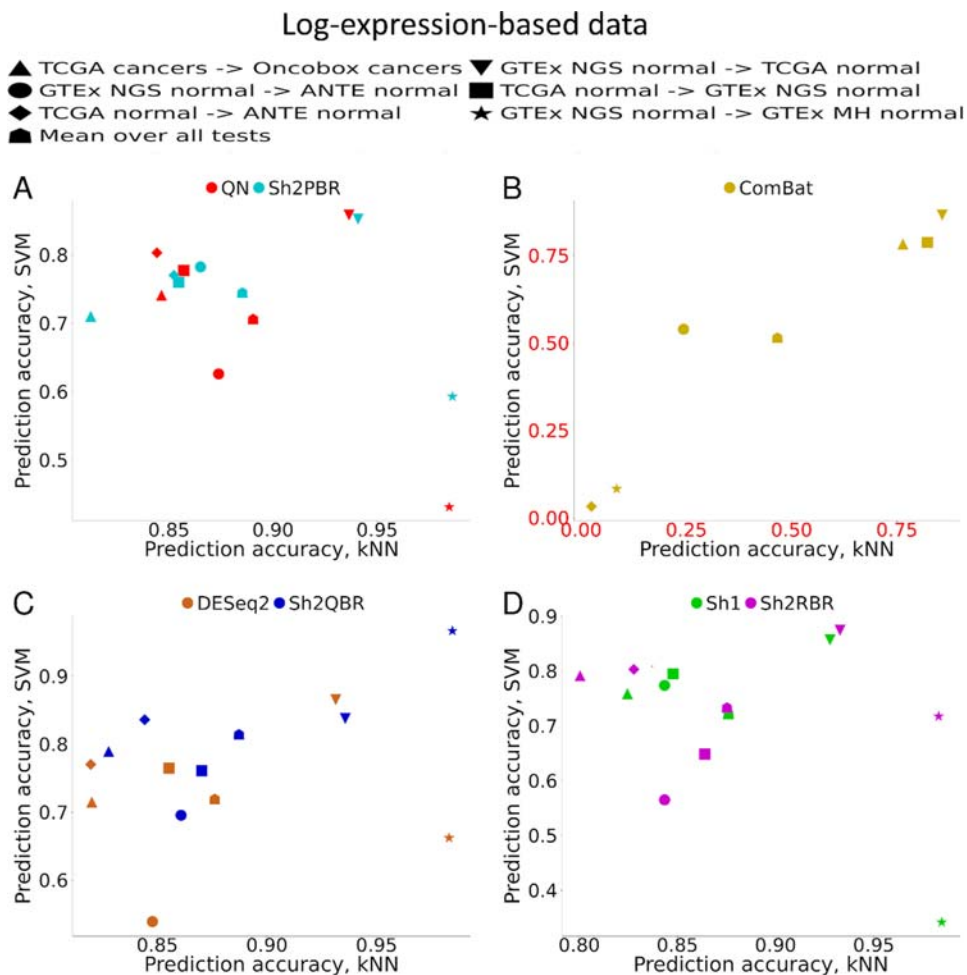


FIGURE 7.3 Total tissue type prediction accuracies calculated for local (kNN) and global (SVM) ML methods for different training and validation datasets. (A) QN, Sh2PBR, (B) ComBat, (C) DESeq2, Sh2QBR, (D) Sh1, Sh2RBR (Borisov et al., 2023b).

In addition, Sh2PBR harmonization mode also showed the optimal performance in the correlation-regression analysis of profiles for the same tissue samples. By averaging the log-expression levels for each gene over all samples of a specific type in the training and validation datasets, we derived the expression level vectors for each tissue type. The distribution of Spearman correlation and linear regression coefficients between the corresponding vectors is shown in Fig. 7.4.

An ideal method for cross-platform harmonization should result in transcriptomic profiles that are as similar as possible for the same samples or tissue types, even when obtained using different platforms or protocols. Consequently, the correlation and linear regression coefficients between the corresponding vectors should be as close as possible to 1 (note that the linear

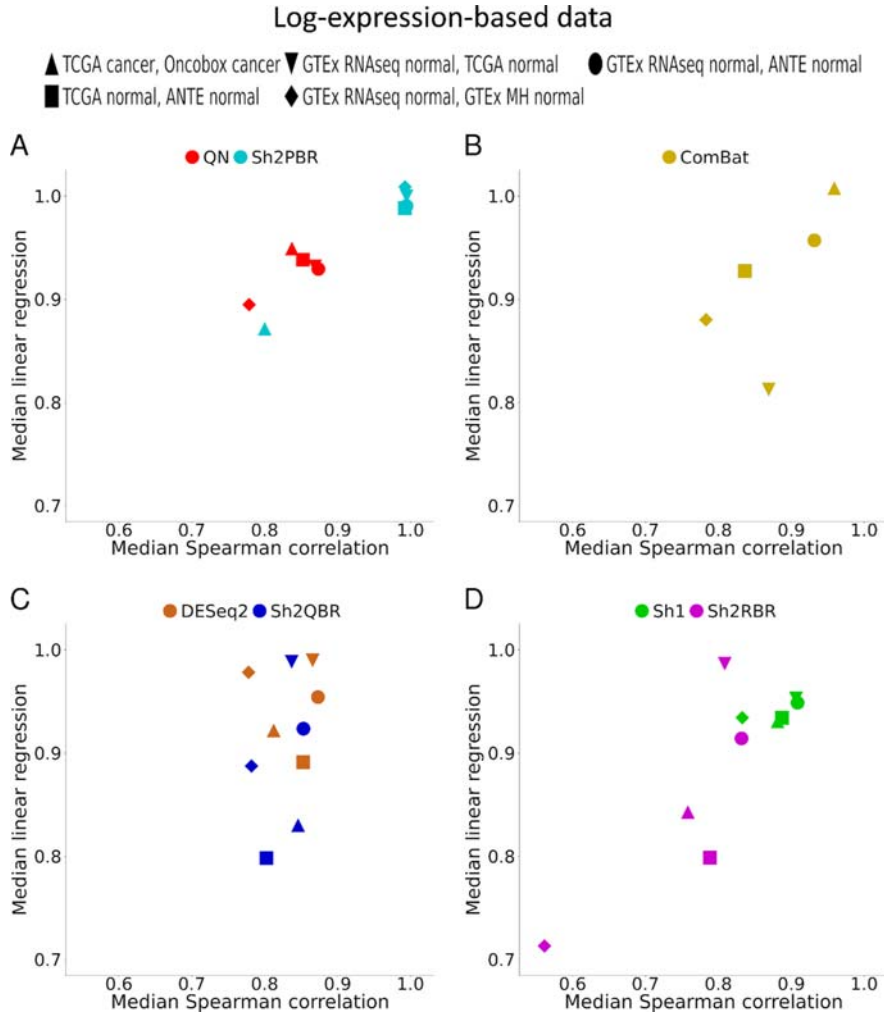


FIGURE 7.4 Median (over all possible tissue types) Spearman correlation and linear regression coefficients between identical tissue types from training and validation datasets. (A) QN, Sh2PBR, (B) ComBat, (C) DESeq2, Sh2QBR, (D) Sh1, Sh2RBR (Borisov et al., 2023b).

regression coefficients may be also lower or greater than 1). We found that the Sh2PBR method results in the correlation and linear regression coefficients very close to 1, thus making it the method of choice for harmonization of human normal transcriptomic profiles (Fig. 7.4).

In brief, we may summarize the following:

- In transfer learning tests, when the ML models were trained on gene expression profiles obtained using one experimental platform and validated on profiles obtained using another platform, the Shambhala modes showed performance that was better or comparable to QN and DESeq2, and superior to ComBat;

- in terms of correlation/linear regression coefficients, the Sh2PBR mode showed the results maximally close to 1;
- the ComBat method, which is designed to eliminate the batch effect intentionally, showed the least performance in the tests with one cancer and four normal tissue datasets.

7.6 Correlation, regression, and sign-change analysis of cancer drug balanced efficiency scores after application of different methods of harmonization

The universal harmonization of gene expression data is of interest not only for proper classification of biosamples but it also has important clinical implications. For example, bioinformatic platform Oncobox is designed for personalized prediction of cancer drug activities using cancer and normal gene expression profiles (Borisov et al., 2020b; Poddubskaya et al., 2019; Tkachev et al., 2020b; Zolotovskaia et al., 2022). It evaluates differential gene expression (reflected by log-fold-change (LFC) values) and identifies altered molecular pathways (reflected by pathway activation levels, PALs) that may serve as the molecular targets for cancer drugs, thus giving BESs for every drug under analysis. Other possible gene expression-based methods of drug efficiency scoring should be also mentioned in this context (Lazar et al., 2015; Solomon et al., 2022). However, finding proper tumor-matching normal tissues is frequently challenging and cannot accommodate for statistically correct differential gene analysis. It is, therefore, important to compare experimental cancer expression with the normal tissue datasets, e.g., published as the part of GTEx, TCGA, and ANTE projects. Such a comparison may require data harmonization, provided that different equipment, reagents, and protocols could be employed for obtaining different datasets.

The Shambhala harmonization has filtered away approximately 60% of protein-coding genes with the lowest expression (Borisov et al., 2023b). However, among all 167 drugs in the Oncobox database, 129 (77%) retained all target genes after harmonization (Borisov et al., 2023b). The enrichment analysis of Gene Ontology terms for the survived versus filtered out genes (Fig. 7.5) shows that the majority of survived genes govern important physiological process, including cell cycle, whereas the majority of rejected genes are related to sensory/olfactory mechanism, and only a minor part of neglected genes deals with mitosis and cancer. It is noteworthy that important genes that distinguish cancer subtypes, e.g., for breast cancer, like ERBB2 (HER2) and ESR1 (ER), and two forms of membrane component of PR, PGRMC1, and PGRMC2, survived shambhalization.

Thus, we assessed the influence of Shambhala harmonization on the BES values for these 129 targeted cancer drugs across a total of 605 experimental RNA sequencing samples from 20 cancer types. We then analyzed the performance of four Shambhala modifications (Sh1, Sh2PBR, Sh2QBR, and Sh2RBR) in a series of tests for correlation, regression, and sign-change assessments.

- First, comparison of BES values, which were obtained using QN for Oncobox cancer and ANTE normal gene expression profiles, which is a routinely used protocol for BES calculation in clinical use (Tkachev et al., 2020b), against those obtained after Shambhala harmonization of cancer and normal profiles (Fig. 7.6).

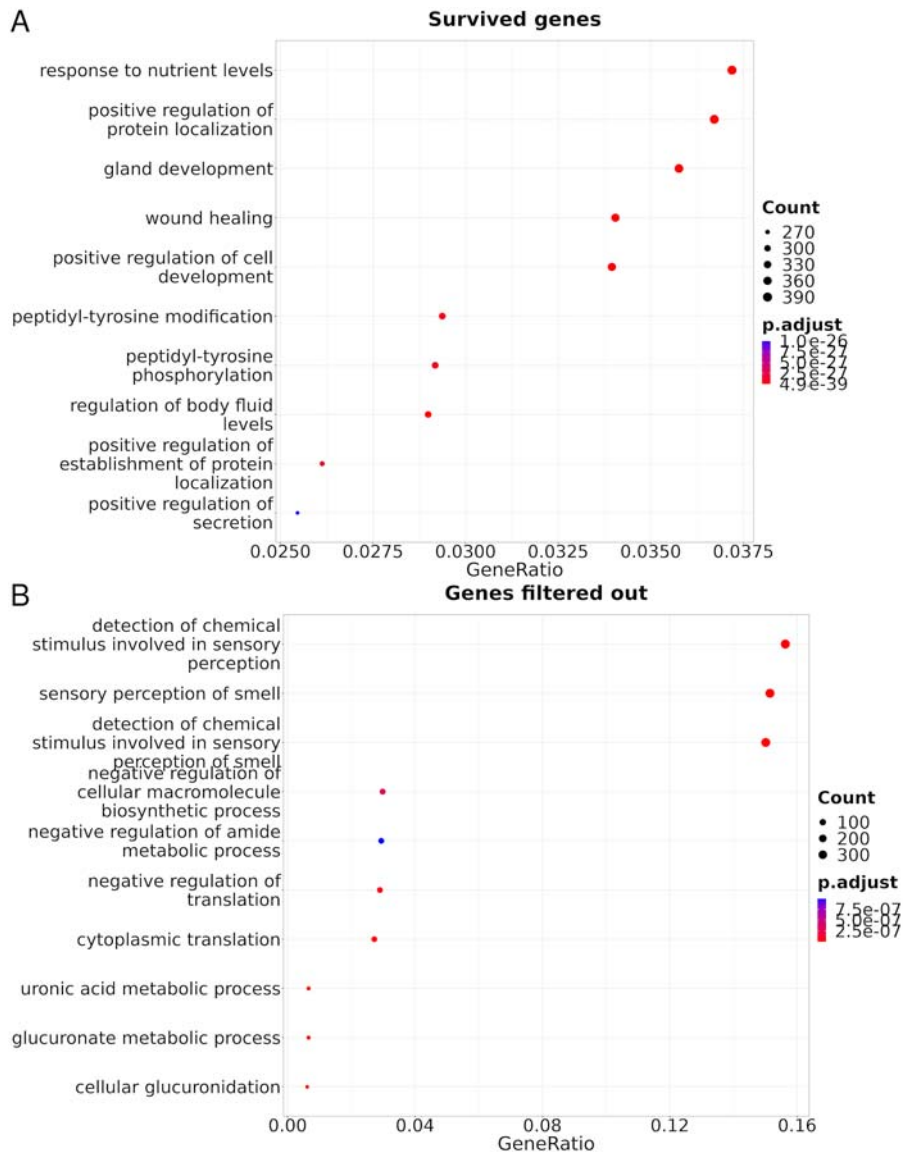


FIGURE 7.5 Gene ontology terms enrichment analysis (Yu et al., 2012) for the genes that survived Shambhala harmonization (A), or that were removed by filtering (B) (Borisov et al., 2023b).

(ii) Second, we compared BES values obtained using ANTE normal gene expression profiles (Suntsova et al., 2019), against those obtained with the corresponding GTEx (GTEx Consortium, 2013) normal profiles (Fig. 7.7).

We found that the BES values for QN and all modes of Shambhala harmonization were strongly correlated in all cancer types under analysis (Fig. 7.6). It can be expected that ideal

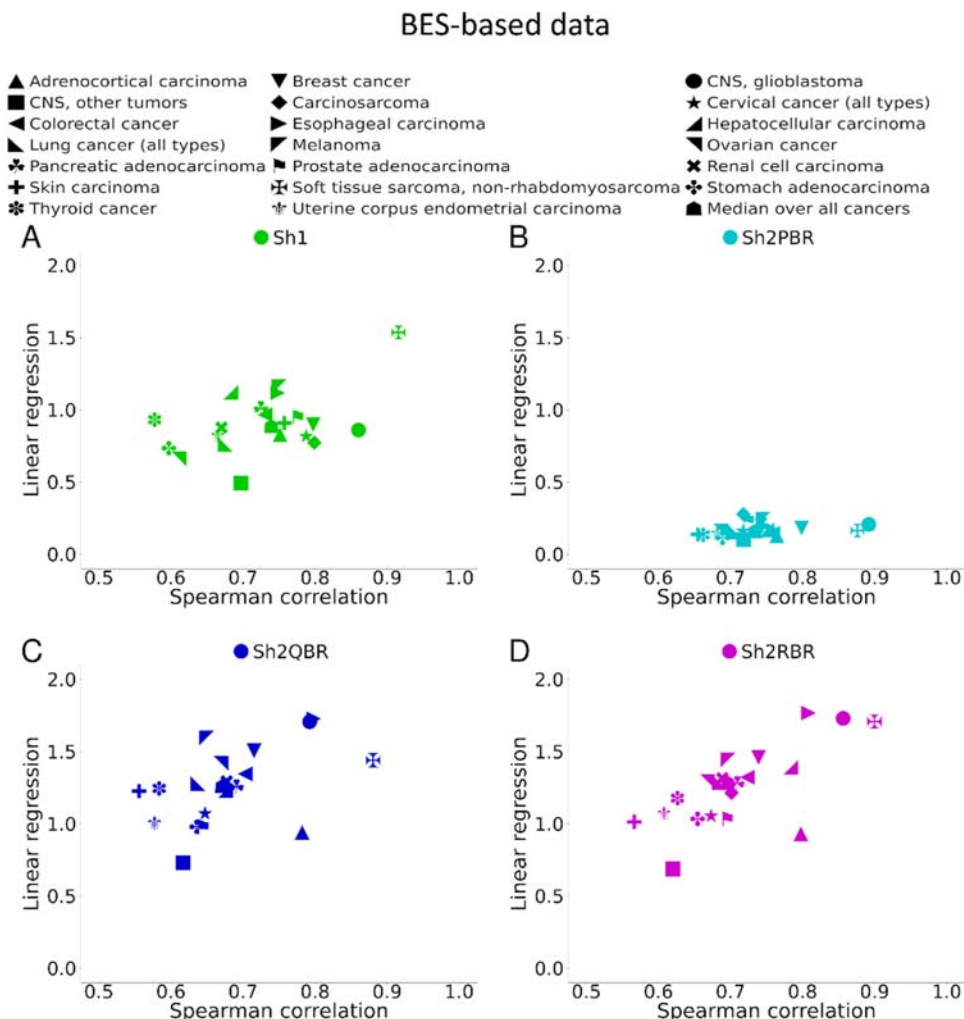


FIGURE 7.6 Spearman correlation and linear regression coefficients between BES values for QN versus Shambhala comparison of 20 cancer types and 129 targeted cancer drugs. (A) Sh1, (B) Sh2PBR, (C) Sh2QBR, (D) Sh2RBR (Borisov et al., 2023b).

data harmonization will keep stable the cancer-to-normal LFC, and subsequent PAL and BES magnitudes, compared to the gold standard normalization methods such as QN or DESeq2. In other words, this should set the linear regression coefficient k close to 1 for the QN versus Shambhala comparison. However, although this ideal figure was more or less true for the Sh1 mode, the best mode in terms of correlation coefficient, Sh2PBR, at the same time exhibited the worst (i.e., the maximally distant from 1) values for the regression coefficient k , with $k < 1$ (Fig. 7.6B). In contrast, other versions of Shambhala-2 (Sh2QBR and Sh2RBR) had $k > 1$ (Fig. 7.6B). We have developed and tested the SH2QBR and Sh2RBR modes as the

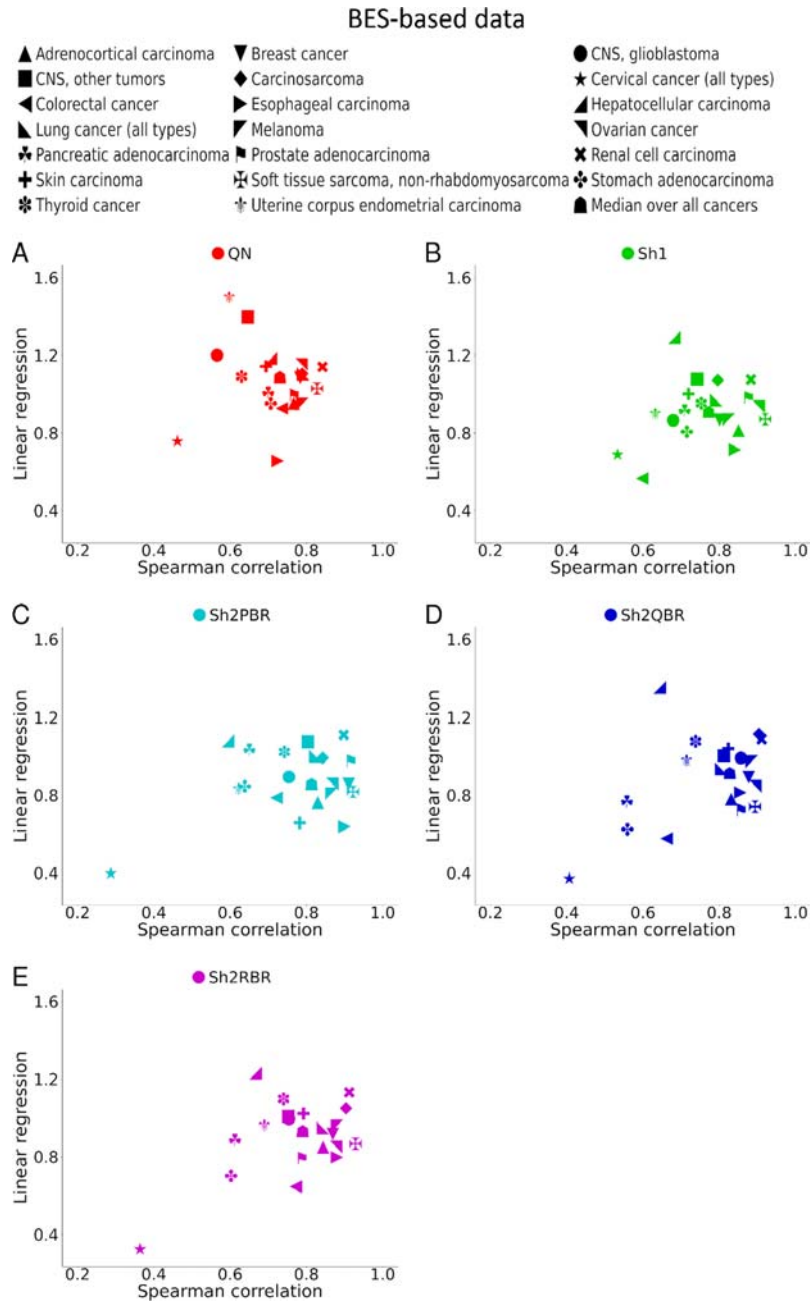


FIGURE 7.7 Spearman correlation and linear regression coefficients between the corresponding BES values for the same cancer drugs calculated with ANTE versus GTEx normal references for 20 cancer types and 129 targeted cancer drugs. (A) QN, (B) Sh1, (C) Sh2PBR, (D) Sh2QBR, (E) Sh2RBR ([Borisov et al., 2023b](#)).

alternatives to the previous mode Sh2PBR (Borisov et al., 2022), since the latter tended to artificially decrease the absolute values of LFC, PAL, and BES metrics.

The values of regression coefficients, k , between BES values calculated using QN and different Shambhala modes, varied significantly from 0.17 till 1.32 (Fig. 7.6). Likewise, the magnitudes of BES varied accordingly. This variation in magnitude prompted us to check whether it can affect the sign of BES. The BES value predicts the ability of a drug to inhibit abnormally activated molecular pathways in individual cancer. The positive/negative sign of the BES values is crucial, as it indicates the potential beneficial/harmful effects of a drug in a particular cancer treatment. We found here that the use of Shambhala harmonization instead of QN has relatively minor effect on the sign of the BES values (Fig. 7.8A). Changes in the sign of BES are meaningful because it indicates whether a drug is potentially helpful against a particular tumor (BES positive), or not (BES negative or zero). With the growth of the significance threshold, the percentage of sign-changing BES values decreased rapidly. The most significant decrease was observed in the comparison between QN and Sh2PBR (Fig. 7.8A).

The division of the Shambhala BES values by the regression coefficient k , we found an increase in the percentage of sign-changing events for the Sh2PBR mode. However, even after this division the Sh2PBR mode showed the best results in terms of BES sign stability: compare the options “as is”, i.e., without correction, and “divided by k ” in Fig. 7.10A. Interestingly, using Shambhala harmonization instead of QN increased correlation between the BES values calculated for Oncobox cancers versus ANTE and GTEx norms (Fig. 7.7). Moreover, using GTEx instead of the ANTE normal samples did not change the BES magnitude much, and all regression coefficients (k) remained close to 1 for the ANTE versus GTEx normal reference comparison (Fig. 7.7).

The proper selection of the P -dataset considerably increases the biologically relevant properties of the harmonized data. In our previous study (Borisov et al., 2022), we tried eight different P -datasets obtained using different MH and NGS profiling methods, and the Affymetrix Human Genome U133A 2.0 Array-based P -dataset apparently showed the best ability for distinguishing the biological nature of samples after harmonization. The relatively good performance of QN in our experiments may be due to the nature of testing gene set including ~ 8000 genes with the greatest expression level, which were included in the P -dataset.

In the Oncobox cancer dataset, the same equipment and protocol were used for RNA sequencing as in the ANTE collection of normal samples; instead, different protocol but the same equipment was used for sequencing in the GTEx project. Thus, technically ANTE norms better correspond to the Oncobox cancers than GTEx norms for the same tissues. However, the number of Oncobox ANTE normal samples is limited, whereas GTEx tissue samples are much more numerous. Fig. 7.7 shows that both correlation and linear regression coefficients between the BES values with Oncobox ANTE and GTEx norms are close to 1 for all Shambhala modes, which perform better than QN. Poor correlation values are noted for cervical cancer when Oncobox ANTE normal samples are substituted with the GTEx samples (Fig. 7.7). This poor correlation may be caused by unusually small number of normal samples (only four samples for the Oncobox ANTE collection, and only 18 samples for the GTEx collection), when the sample heterogeneity may play a crucial role.

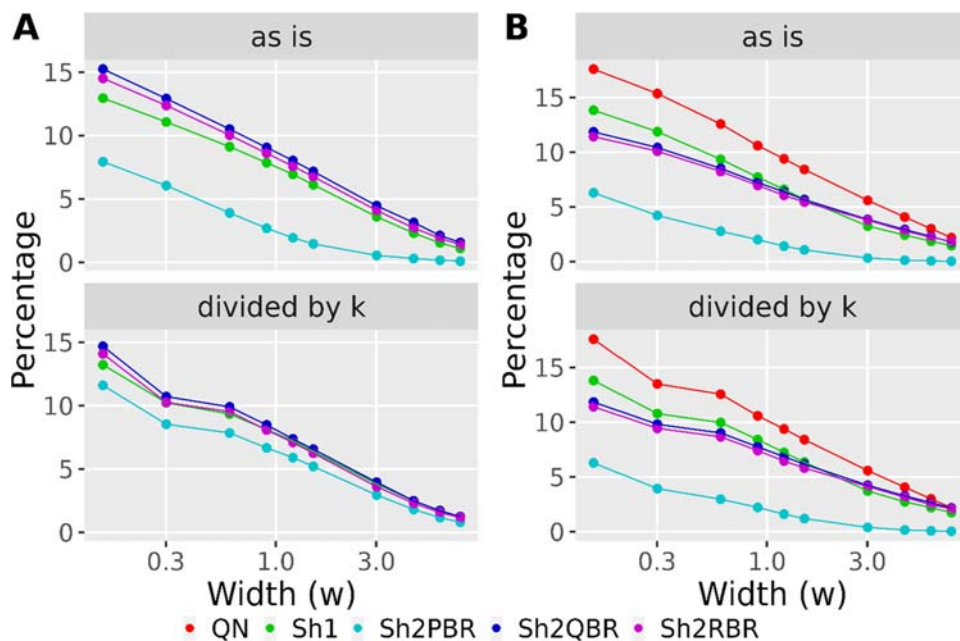


FIGURE 7.8 Percentage of BES values, which change their sign when using Shambhala harmonization instead of QN (A), and when using GTEX instead of the ANTE normal references (B), as a function of width (w) for the sign-changing significance threshold in all 605 experimental Oncobox samples of 20 cancer types, and 129 anticancer target drugs. Mode “as is” is given for BES values without correction. Mode “divided by k ” is given for BES values divided by the corresponding linear regression coefficients (see Fig. 7.3 for panel A, and Fig. 7.4 for panel B). The width (w) was measured in terms of median absolute values of BES for all cancer cases (Borisov et al., 2023b).

We, therefore, conclude that our tests with the case-to-control LFC, PAL, and BES metrics showed that the Sh2PBR mode had the best performance in terms of close to 1 correlation and regression coefficients, and the minimal percentage of sign-changing LFC/PAL/BES values (Borisov et al., 2023b).

7.7 Retention of biological properties after uniformly shaped harmonization

Besides many statistical performance indicators such as correlation and regression coefficients, sign stability rates, and ML accuracy, it is also important whether the harmonization retains general biological characteristics of biosamples. Otherwise harmonization output will have little sense even when the statistical metrics looks acceptable. One such simple test for retention of biological significance can be done with the sex-specific gene expression. Out of 848 X-chromosome genes, 285 survived Shambhala harmonization, and none survived among the 47 Y-chromosome genes (Borisov et al., 2023b). Using expression of these survivor X-linked genes as the biomarkers, we performed a principal component analysis assay to check for the presence of distinct clusters formed by the male and female patient biosamples

from the Oncobox database (for 202 male and 357 female patients). It can be seen from Fig. 7.9 that all Shambhala modes retained sex-specific gene expression pattern comparable to QN, and better than that obtained for DESeq2.

We also considered the differential gene patterns among different breast cancer molecular subtypes. Although we expect no essential batch effect in the Oncobox data, the overall expression pattern seems tangled. In terms of clustering dendrograms, the hormone-

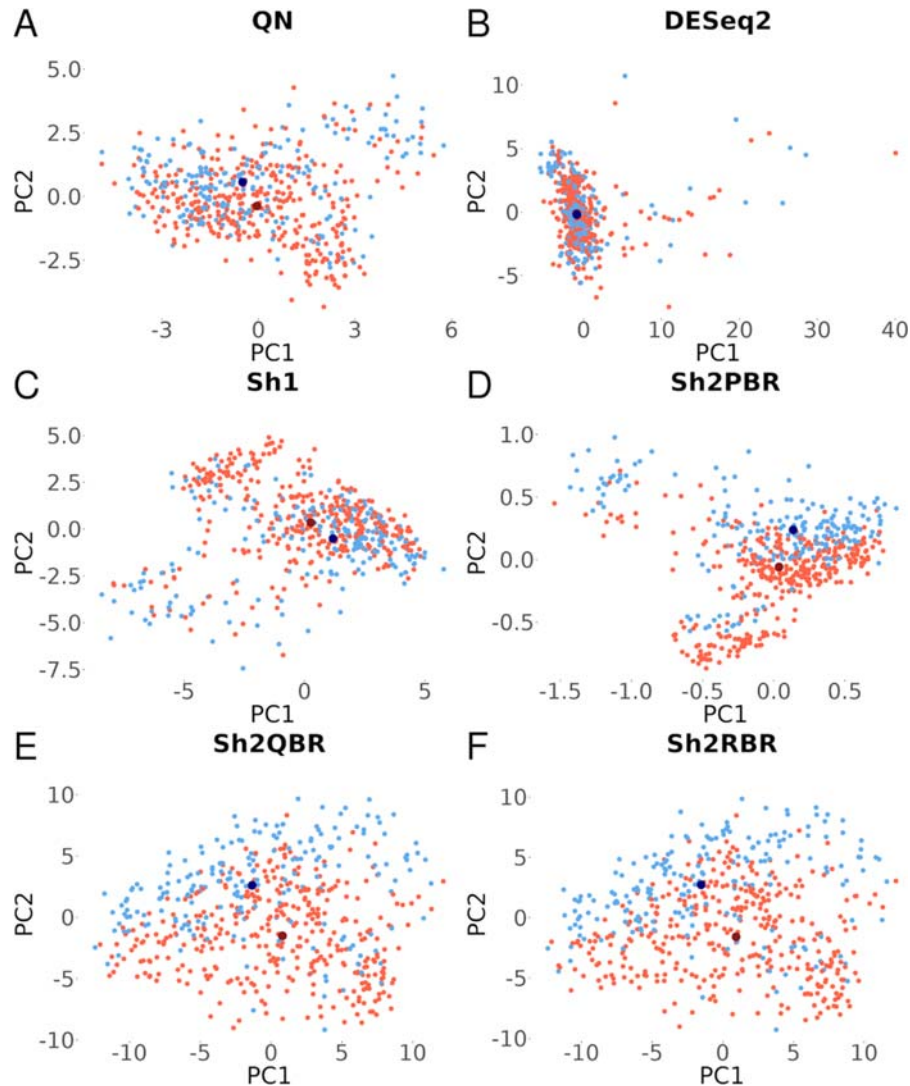


FIGURE 7.9 Principle component analysis of the 285 genes located on X chromosome, which survived Shambhala harmonization. Red and blue dots correspond to female and male cancer patients, respectively. Dark dots represent the median values. (A) QN, (B) DESeq2, (C) Sh1, (D) Sh2PBR, (E) Sh2QBR, (F) Sh2RBR (Borisov et al., 2023b).

dependent subtypes (ESR1 and PGR-positive), HER2-positive, and triple negative cancer samples are mixed together across all normalization/harmonization methods (Borisov et al., 2023b). The standard pipelines for differential gene expression analysis (Kuznetsova et al., 2021) seem inappropriate, as Shambhala affects the absolute values of LFC. However, Shambhala showed strong ability to retain ROC AUC¹ metrics between these tumor subtypes. Here, we obtained pools of marker genes by calculating AUC (with the threshold of AUC >0.8) for two pairwise comparisons: (i) HER2-positive versus hormone-dependent BC; (ii) triple negative versus HER2-positive BC. For triple negative versus HER2-positive, and for HER2-positive versus hormone-dependent breast cancer, the Sh2QBR and Sh2RBR methods introduced numerous artifacts with the marker genes that were absent for QN. Interestingly, DESeq2 also identified many genes with AUC >0.8 for HER2-positive versus hormone-dependent comparison that were absent for QN, Sh2PBR, and Sh1 (Fig. 7.10).

Thus, our tests with the expression of the X-chromosome genes in male and female cancer patients, as well as with marker genes for the hormone-dependent, HER2-positive, and triple-negative breast cancers, confirmed the superior performance of the Sh1 and Sh2PBR modes over the other methods tested.

7.8 Validation of Shambhala-1/2 protocol on sc-mRNA-seq profiles

Although the Shambhala methods were first suggested for the cross-platform harmonization of bulk mRNA-seq and MH profiles, they appeared useful for harmonization of single-cell mRNA sequencing (sc-seq) also.

Importantly, these methods demonstrated the ability to process even very large numbers of profiles, where standard methods for variance reduction become impractical due to the enormous RAM capacity and the computational time required for calculations.

We have performed a harmonization/variance reduction test with more than 200,000 single-cell profiles derived from retinal pigment epithelium (RPE1) cell lines (Replegle et al., 2022). During this large genotype-to-phenotype mapping study, more than 2000 different genes were knocked-out many dozen times, resulting in more than 200,000 single-cell profiles measured on the Illumina NovaSeq 6000 device. Generally, the sc-profiles have very strong variance-to-mean correlation (Fig. 7.11A), which cannot be destroyed even after the application of Markov affinity-based graph imputation of cells (MAGIC) procedure (Andrews and Hemberg, 2018; Hou et al., 2020; Luo et al., 2020; Su et al., 2014; Xu et al., 2021), a standard protocol for preprocessing of sc-seq data (Fig. 7.11B).

The analysis of noise-generating knockouts (KO) using the sc-seq profiles generally involves variance reduction methods, such as removal of unwanted variance (RUV) (Hung, 2019; Molania et al., 2023; Salim et al., 2022). As shown in Fig. 7.11, panels B and F, selecting three profiles for each KO produces only ~6000 profiles in the sampled dataset. However,

¹ The ROC (receiver–operator curve) is a widely used graphical plot that illustrates the diagnostic ability of a binary classifier system as its discrimination threshold is varied. The ROC is created by plotting the true positive rate (TPR) against the false positive rate (FPR) at various threshold settings. The area under the ROC curve, called ROC AUC, or simply AUC, is routinely employed for the assessment of any classifier quality.

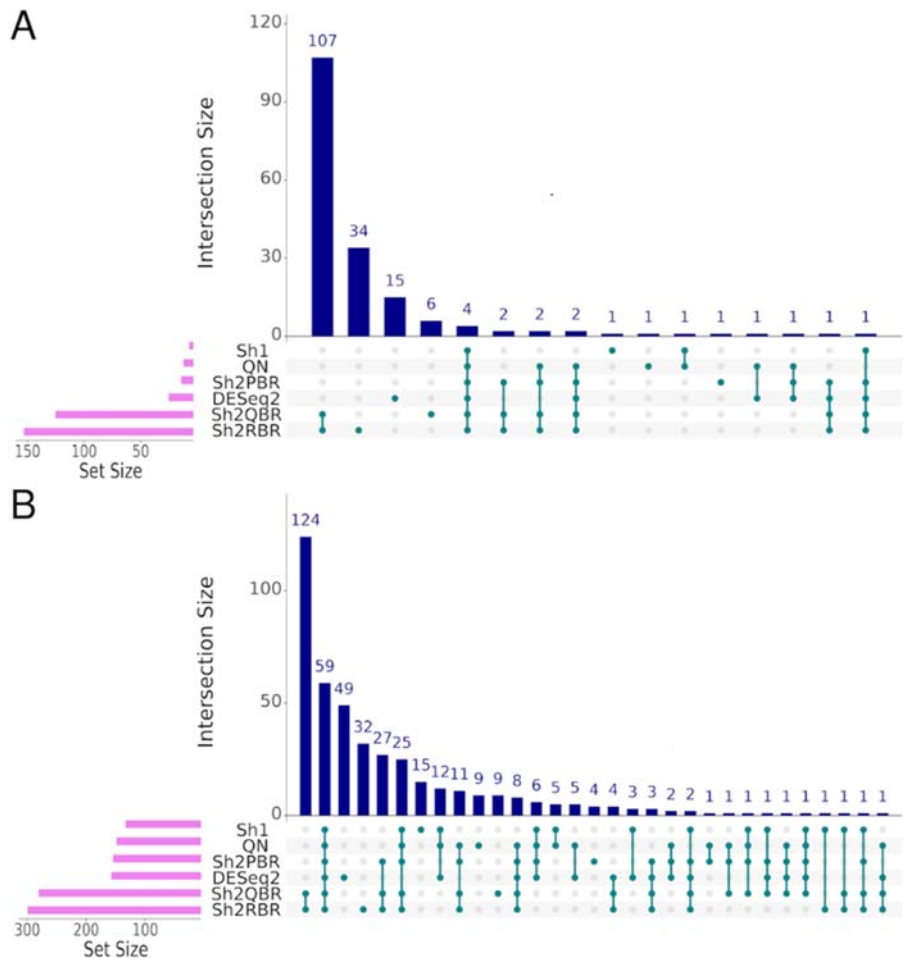


FIGURE 7.10 UpSet intersection diagrams (Conway et al., 2017) of marker genes (AUC >0.80) for the comparison of breast cancer (BC) expression profiles of different molecular subtypes. (A) HER2-positive versus hormone-dependent BC. (B) Triple negative versus HER2-positive BC. Along the x-axis of the UpSet diagrams, all observed intersections of the analyzed sets are represented. The intersections are marked with turquoise dots, with the rows of the dot matrix corresponding to different sets, and columns showing the intersection combinations (Borisov et al., 2023b). UpSet diagrams were proposed as more informative alternative to traditional Venn diagrams for the analysis of more than four intersecting datasets (Lex et al., 2014).

this was the limit of RUV’s abilities for noise reduction, even at the Amazon AWS virtual machine with 128 GB of RAM and 64 CPUs. For the whole dataset with ~200,000 profiles, RUV was completely unable to handle these extremely Big Data due to the quadratic time and computational complexity.² Although for processing relatively small numbers of

² In algorithm complexity theory, time and RAM capacity required for the completion of calculations are called time and space complexity, respectively. This complexity may be linear, log-linear, quadratic, etc., depending on a particular parameter of a model, e.g., the number of profiles in a dataset.

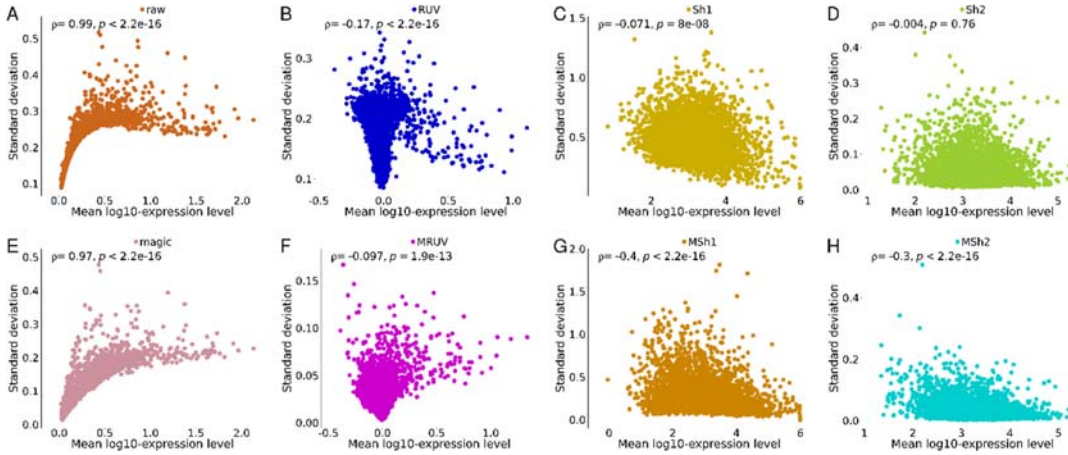


FIGURE 7.11 Variance-to-mean plots for RPE1 sc-profiles (Replogle et al., 2022) (for each KO gene, three profiles were randomly selected, making ~ 6000 profiles in a subsample): (A) raw data, (B) RUV noise reduction, (C) Shambhala-1, (D) Sh2PBR, (E) MAGIC imputation, (F) RUV after MAGIC, (G) Shambhala-1 after MAGIC, (H) Sh2PBR after MAGIC.

transcriptomic profiles Shambhala-1/2 is slower than most other methods due to separate processing of each profile, it can become preferable for really Big Data Shambhalain terms of both computational time and memory capacity. This is because it has a guaranteed linear time and space computational complexity. At the Amazon AWS virtual machine with 128 GB of RAM and 64 CPUs, the complete Shambhalization of a whole PRE1 sc-seq dataset (Replogle et al., 2022) took about 7 days and produced results similar to those for the 6000-profile subsample (Fig. 7.11, panels G, D, G, H).

Moreover, for handling the Big sc-seq Data Shambhala-2 exhibited one more advantage over RUV: it produced better articulation of distinct cell-cycle phases. For the ~ 6000 profile subsamples, we annotated profiles in terms of cell-cycle phase articulation. This annotation uses marker gene list for cell-cycle phases from the Python library Perturbseq.³ For each gene, we first performed central z-scoring of log-expression level through all profiles, obtaining the z_g values for each gene. For each profile, we then calculated the mean z-scored log-expression levels among the cell cycle phase marker genes, arriving at z_i values, where i runs through different cell-cycle phases. The cell-cycle stage flag assigned for each profile corresponds to phase i with maximal z_i .

The profiles annotated with the most articulated cell cycle phase were then analyzed using the WM-metric for clustering dendrograms. We obtained the highest WM value for the Shambhala-2 method preceded by the MAGIC imputation (Fig. 7.12H).

Thus, the Shambhala-2 method was also the best for unwanted variance reduction in the sc-seq Big Data (200,000 profiles), and highlighting the cell cycle phases.

³ Perturbseq Python library can be found at: https://github.com/thomasmaxwellnorman/Perturbseq_GI/blob/master/perturbseq/cell_cycle.py

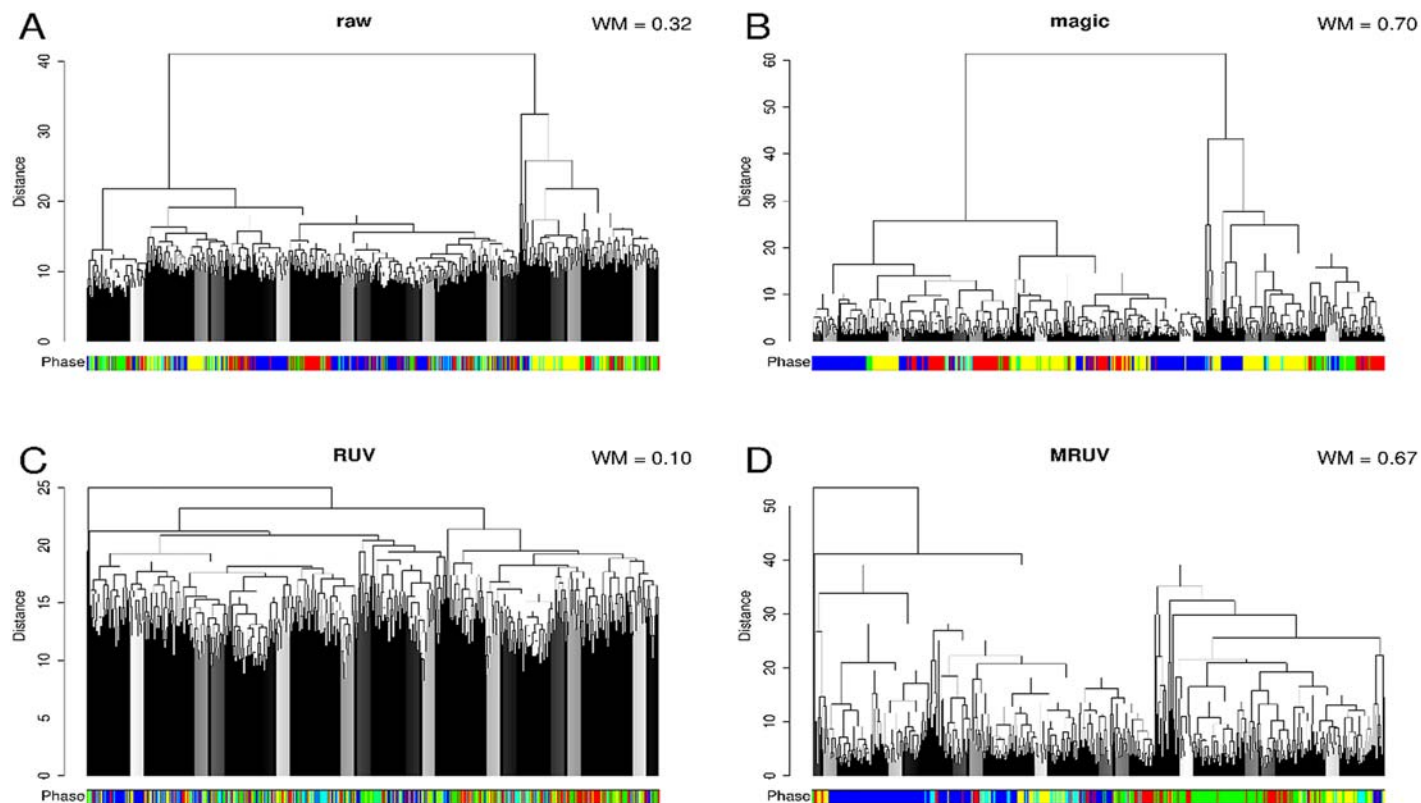


FIGURE 7.12 Recognition of cell cycle phases (red: G1/S; yellow: G2/M; green: S; cyan: M; blue: M/G1) using the WM-metric (Borisov et al., 2022, 2023b; Zolotovskaia et al., 2020) (in the top right corner) for dendrograms for 6000 RPE1 sc-seq profiles: (A) raw data, (B) MAGIC imputation, (C) RUV noise reduction, (D) RUV after MAGIC, (E) Shambhala-1, (F) Shambhala-1 after MAGIC, (G) Sh2PBR; (H) Sh2PBR after MAGIC.

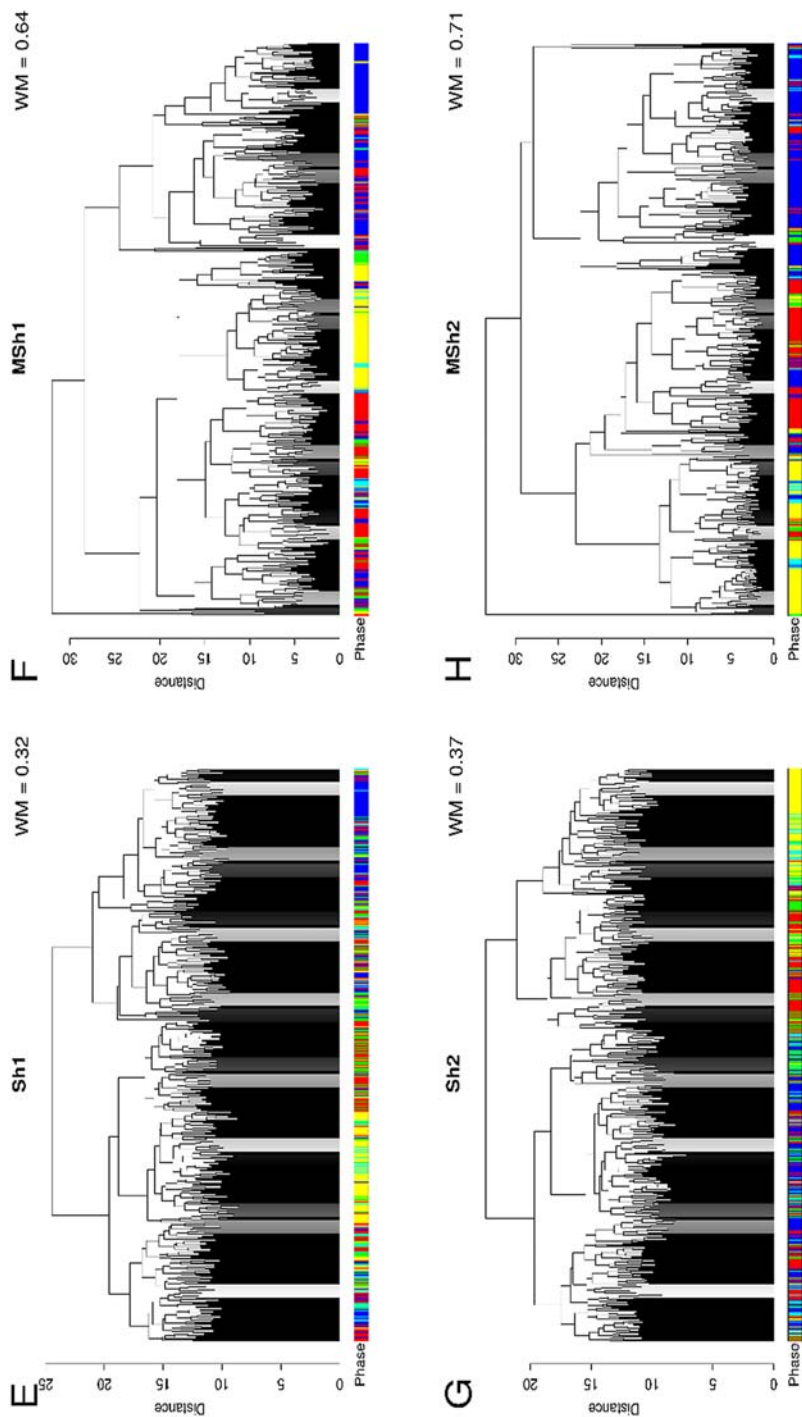


FIGURE 7.12 (Continued.)

7.9 Discussion

Dozens of methods for cross-platform normalization have been developed for the analysis of both microarray hybridization and RNA sequencing gene expression data, yet none of them is currently recognized as an overall gold standard (Borisov and Buzdin, 2022). The majority of these methods returns output data in a flexible format (Franks et al., 2018; Huang et al., 2022; Maleknia et al., 2020; Piccolo et al., 2013; Tang et al., 2021; Thompson et al., 2016; Zhang et al., 2020) that requires recalculation of all expression samples every time upon the addition of new gene expression profiles. In search for the universal output format, we developed a novel approach termed Shambhala for uniformly shaped cross-platform harmonization of gene expression data (Borisov et al., 2019, 2022; Borisov and Buzdin, 2022). The key feature of Shambhala is the one-by-one conversion of each individual profile into the universal shape of the reference definitive dataset independently from the other profiles. This not only creates the basis for unlimited number of further updates of the gene expression bank(s), but also makes it possible to combine together any number of expression datasets. Shambhala allows adding new samples to previous normalized/harmonized dataset without the need for renormalizing them, with no currently known limitations for the number of merged datasets and number of samples in such datasets.

In our opinion, the current versions of Shambhala have the following major limitations: limited repertoire of normalized human genes ($\sim 8,000$, or $\sim 40\%$ of all protein-coding genes) and higher calculation costs per individual profile due to the algorithm complexity. However, Shambhala has the following advantage over other normalization methods: its output is returned in a universal format comparable with all other “shambhalized” human profiles obtained using any experimental platform or protocol. This requires no recalculation of the whole dataset upon the addition of new sample(s). In the current version of Shambhala, the maximum amplification of signal-to-noise ratio was prioritized whereas it resulted in a reduced spectrum of genes in the harmonized output. Thus, we added reliability expression filters (Borisov et al., 2019) and the selection of the most highly expressed genes during intersection with the auxiliary datasets (Borisov et al., 2022). The resulting reduced gene set, however, includes 73%–89% of genes participating in molecular pathways (Borisov et al., 2023b), depending on the pathway database (Croft et al., 2014; Nishimura, 2001; Schaefer et al., 2009; Zolotovskaia et al., 2022) and $\sim 90\%$ of molecular targets of targeted cancer therapeutics which makes such an analysis meaningful for cancer research. However, developing the next versions of Shambhala enabling the high-quality transformation of a greater proportion (ideally all) of human genes, and also non-human genes for other model objects will be a matter of our further studies.

Our previous analysis (Borisov et al., 2022) has demonstrated the strong capacity of the cubic transformation-based Shambhala method (Shambhala-2) to eliminate the platform bias from the clustering dendrograms of gene expression profiles, which assured grouping of the samples according to their biological origin. Here, we further assessed the performance of four modifications of Shambhala by applying 12 criteria that characterize the retention of

biological functional properties and differential gene expression patterns in thousands of samples from various human tissue types.

Ideally, harmonization of gene expression should provide (Borisov et al., 2023b):

- Clustering of samples according to their biological features, and not according to technical factors like platform/protocol used;
- close to one correlation coefficients for gene expression values in comparison with the initial datasets;
- close to one linear regression coefficients for relative gene expression values and their derivatives like pathway activation levels (PAL) and drug efficiency scores (BES) in comparison with the initial datasets;
- stability of sign for logarithmic relative gene expression values like fold-change and their derivatives like drug efficiency score in comparison with the initial datasets.
- conservation or “common sense” biological properties, such as expression levels of sex-related genes, or marker genes between disease subtypes.

In addition, taking into account Big Data analytic approaches, another desirable feature is data suitability for various ML methods, *e.g.*, local and global. An ideal harmonization of gene expression data has to generate similar molecular profiles for the samples of similar biological origin, even when obtained using different equipment and protocols. This means that the ML process trained on the harmonized data obtained using one platform and validated on the harmonized data obtained using another platform should show high overall accuracy. Likewise, the profiles of similar biological origin, but obtained using different equipment/protocols, should be strongly correlated.

We compared the performance of four available versions of Shambhala with the gold standard flexible output methods QN and DESeq2 according to 12 analytic criteria reflecting the above consideration (Borisov et al., 2023b). For eight out of these 12 criteria, the Sh2PBR mode was better than any other harmonization methods. For three other criteria, the Sh2PBR method was atleast comparable to any other methods under consideration. Only for the one criterion, the method Sh2PBR showed unusually low linear regression coefficient $k \sim 0.20 \div 0.25$, which indicates that Sh2PBR causes four-to-five-fold decrease of case-to-control log-fold changes in gene expression levels (Fig. 7.6B). However, this proportional variation of case-to-control log-fold-changes does not perturb the correlation coefficients, order, and expression ranks of the individual genes and molecular pathways, and also of the predicted drug efficiency scores. Moreover, after Sh2PRB harmonization, the percentage of sign-changed BES values remained lower than all other harmonization methods by approximately 20% (Fig. 7.8A and B).

In Table 7.2, we summarized the recommendations for the use of different normalization/harmonization methods. Overall, our correlation, regression, sign-change analysis, unwanted variance reduction trial, and cell-cycle phase articulation test demonstrate the best results of the Sh2PBR version of Shambhala. We, therefore, suggest that Sh2PBR can be considered as the method of choice for harmonization of various types of human gene expression data.

TABLE 7.2 Recommendations for applications of gene expression normalization/harmonization methods (Borisov et al., 2023b).

References	Method	Mathematical Principle	Algorithmic Comprehensiveness	Advantages	Shortcomings
Bolstad et al. (2003)	Quantile normalization (QN)	Ranking the expression levels of different genes within each profile and setting the expression level of each gene to the mean value (over all profiles) for the respective rank	Relatively easy	Gold standard method for intraplatform normalization of the MH data	Cannot be used for cross-platform harmonization of the MH data; Requires recalculation of all gene expression-based values after the addition of new samples
Love et al. (2014)	Differential gene expression in sequencing 2 (DESeq2)	Transform based on the negative binomial distribution	Relatively easy	Gold standard for intraplatform normalization of RNAseq data	Requires recalculation of all gene expression-based values after the addition of new samples
Borisov et al. (2019)	Shambhala-1 (linear Shambhala)	Uniformly shaped harmonization based on the XPN (Shabalin et al., 2008) method.	Not so easy to understand	Suitable for harmonization of unlimited number of datasets of any size, for both MH and RNAseq data or their combinations; does not require recalculation of gene expression-based values after addition of new samples. Suitable for Big data.	Resource-demanding. Reduces the number of protein-coding genes in the harmonized output down to ~8000 items.
Borisov et al. (2022)	Shambhala-2 (cubic Shambhala)	Uniformly shaped harmonization based on the CuBlock (Junet et al., 2021) method.	Not so easy to understand	Suitable for harmonization of unlimited number of datasets of any size, for both MH and RNAseq data or their combinations; does not require recalculation of gene expression-based values after addition of new samples. Suitable for Big data.	Resource-demanding. Reduces the number of protein-coding genes in the harmonized output down to ~8000 items.

Abbreviations

ANTE – Atlas of Normal Tissue Expression
AUC – Area under curve
AUROC – Area under Receiver–Operator Curve
BC – Breast cancer
BES – Balanced efficiency score
CPU – Central processing unit
CuBlock – Cubic blocks
DES – Drug efficiency score
DESeq2 – Differential Gene Expression Analysis in Sequencing 2
ER – estrogen receptor
GTE_x – Genotype-tissue expression
kNN – *k* Nearest Neighbors
KO – knockout
LFC – log-fold-change
MAGIC – Markov Affinity-based Graph Imputation of Cells
MH – Microarray hybridization
NGS – Next-generation sequencing
PAL – Pathway activation level
PR – progesterone receptor
QN – Quantile normalization
RAM – Random-access memory
RNAseq – RNA sequencing
ROC – Receiver–operator curve
RPE – retinal pigment epithelia
RUV – Removal of Unwanted Variance
sc-seq – single-cell sequencing
Sh1 – Shambhala-1
Sh2 – Shambhala-2
Sh2PBR – Shambhala-2 with P-Based Rescaling
Sh2QBR – Shambhala-2 with Q-Based Rescaling
Sh2RBR – Shambhala-2 with R-Based Rescaling
SVM – Support vector machines
TCGA – The Cancer Genome Atlas
TDM – Training distribution machine
UPC – Universal exPression Code
WM – Watermelon multisection
XPN – Cross-platform normalization

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